CS 223 Digital Design Laboratory Assignment 1

Digital Building Blocks

Lab Dates and Times

Section 1: March 4, 2026 Wed. 08:30-12:20
Section 2: March 2, 2026 Mon. 13:30-17:20
Section 3: March 5, 2026 Thur. 08:30-12:20
Section 4: March 3, 2026 Tue. 08:30-12:20

Location: EA-Z04 (in the EA building, straight ahead past the elevators)
Groups: Each student will do the lab individually. Group size = 1

Preliminary Report Due: March 1, 2026 at 23:59
Lab Report Due: Immediately after the Lab session.

Important Notes:

- Include your code as plain text, not image!
- You should not upload reports with handwriting. Use computers to write your reports, for circuit schematics, you can use online tools such as Draw.io or Vivado schematics.
- Name your reports in the format: `Name_Surname_ID.pdf`
- Your report should have a cover page including: course code, course name and section, the number of the lab, your name-surname, student ID, date.
- Upload your report in the PDF format.
- **Mentioned criteria above are mandatory, if any of them are not followed, the preliminary work will not be accepted.**

Preliminary Work [40]

Part A: Basic Components [10 pts]

- **[2]** Design a 4-to-1 multiplexer by interconnecting three 2-to-1 multiplexers using hierarchical modeling approach, provide the design with schematic diagram.
- **[2]** Design a 16-to-1 multiplexer by interconnecting five 4-to-1 multiplexers using hierarchical modeling approach, provide the design with schematic diagram.
- **[2]** Provide the graphical symbol, logic diagram, and truth table of a 2-to-4 decoder. Include the "Enable" signal.
- **[2]** Design a 3-to-8 decoder using 2-to-4 decoders and minimal amount of gates, provide the design with schematic diagram. You can draw 2-to-4 decoders as a block. Include the "Enable" signal.
- **[2]** Write a hierarchical SystemVerilog module for 3-to-8 decoder using 2-to-4 decoders, and include a testbench for it.

Part B: Draw the circuit schematic described computational unit. [15 pts]

A new computational unit needs to be designed that can perform both addition and subtraction on 4-bit binary numbers. However, the designer faces a challenge in efficiently handling both operations within a single circuit.

Problem Statement: You need to design a circuit that can:

- Takes two 4-bit binary numbers as inputs: $A = a_3a_2a_1a_0$ and $B = b_3b_2b_1b_0$.
- Use a control signal to determine whether to perform addition or subtraction.

- Efficiently handle subtraction using a suitable mathematical approach.
- Provide a correct 4-bit output along with an additional status signal.

How can you achieve this using only full adders and logic gates, without using any full subtractor? Think carefully about how subtraction works in binary arithmetic and design a solution that meets all the requirements.

Hint-1: Think about how computers perform subtraction using only addition. How can you represent B in a way that allows you to add it to A and still get the correct subtraction result?

Hint-2: If you wanted to selectively invert B depending on whether you are adding or subtracting, what simple logic gate could help you achieve this? Also, consider how the initial carry-in might influence the operation.

Part C: Logic Function. [15 pts]

- Design the schematic and write a SystemVerilog module that implements.

$$F(A, B, C, D, E) = \sum(0, 1, 2, 3, 4, 10, 11, 16, 17, 18, 19, 25, 28, 29)$$

- The implementation must use exactly one 8-to-1 multiplexer and one 2-to-4 decoder. No other logic gates (AND, OR, XOR, etc.) are allowed. Complements of input signals may be used.

Lab Work [60]

1- Implement Multiplexer on FPGA [15 pts]

A multiplexer ("MUX") is a fundamental building block in digital design. An M-to-1 MUX selects one of M data inputs and forwards it to a single output, based on the values of its select lines. Larger multiplexers can be constructed hierarchically by interconnecting smaller ones, which you will practice in this part.

- Write structural SystemVerilog code for an 16-to-1 multiplexer constructed from five 4-to-1 multiplexer blocks.
- **[5]** Using System Verilog testbench code, verify in simulation that your 16-to-1 multiplexer operates correctly.
- **[10]** Using your assigned BASYS-3 switches and LEDs, test the multiplexer. Once you verify correct operation, demonstrate the hardware results to the TA and be prepared to answer related questions.

2- Implement Decoder on FPGA [15 pts]

A decoder is a fundamental building block in digital design. A decoder activates exactly one output corresponding to the binary value of its input signals, while all other outputs remain inactive. When an enable signal is used, all outputs are forced to zero when the decoder is disabled.

- Write behavioral SystemVerilog code for an enabled 3-to-8 decoder using Boolean equations with continuous assignments.
- **[5]** Using SystemVerilog testbench code, verify in simulation that your 3-to-8 decoder operates correctly. Ensure that the port order in the module definition and its instantiation match exactly.
- **[10]** Using your assigned BASYS-3 switches and LEDs, test the decoder on the FPGA. Once you verify correct operation, demonstrate the hardware results to the TA and be prepared to answer related questions.

3- Implement Preliminary Design Part B on FPGA[15 pts]

In this part, you will implement the computational unit that you designed in Preliminary Work Part B on the BASYS-3 FPGA. A standalone BASYS-3 board connected to your computer is sufficient for this lab; the Beti board is not required. Use the available switches and LEDs on the BASYS-3 board for testing.

- Write a SystemVerilog module that implements your computational unit design.
- **[5]** Using SystemVerilog testbench code, verify in simulation that your design operates correctly.

- **[10]** Using the switches and LEDs assigned in your constraint file (`.xdc`), test the design on the FPGA. Once you verify correct operation, demonstrate the physical implementation to the TA and be prepared to answer related questions.

4- Implement Logic Function on FPGA [15 pts]

In this part, you will implement the logic function designed in the preliminary work using a single 8-to-1 multiplexer and a single 2-to-4 decoder. The use of inverted input signals is allowed.

$$F(A, B, C, D, E) = \sum(0, 1, 2, 3, 4, 10, 11, 16, 17, 18, 19, 25, 28, 29)$$

- **[5]** Using System Verilog testbench code, verify in simulation that your design operates correctly.
- **[10]** Using switches as inputs and an LED as output, implement the circuit on the BASYS-3 FPGA. Once you verify correct operation, demonstrate the working design to the TA and be prepared to answer related questions.

Lab Policies

1. Clean up your lab station, and return all the parts, wires, the Beti trainer board, etc. Leave your lab workstation for others the way you would like to find it.
2. There are three computers in each row in the lab. Don't use middle computers, unless you are allowed by lab coordinator.
3. You borrow a lab-board containing the development board, connectors, etc. in the beginning. The lab coordinator takes your signature. When you are done, return it to his/her, otherwise you will be responsible and lose points.
4. Each lab-board has a number. You must always use the same board throughout the semester.
5. You must be in the lab, working on the lab, from the time lab starts until you finish and leave. (bathroom and snack breaks are the exception to this rule). Absence from the lab, at any time, is counted as absence from the whole lab that day.
6. No cell phone usage during lab. Tell friends not to call during the lab hours–you are busy learning how digital circuits work!
7. Internet usage is permitted only to lab-related technical sites. No Facebook, Twitter, email, news, video games, etc–you are busy learning how digital circuits work !
8. If you come to lab later than 30 minutes, you will lose that session completely.
9. When you are done, DO NOT return IC parts into the IC boxes where you've taken them first. Just put them inside your Lab-board box. Lab coordinator will check and return them later.